

Name: Bryan Tiamson

Introduction

Melissa and Doug® released a new educational toy product called Build a Tower and need a plan to manufacture one of its parts: the Tower Topper (Figure 1). This manufacturing plan addresses the quantity and method of manufacturing across the first 18 months after the product release, in order to meet the demand shown by the projected sales (Appendix A) without breaking the just in time policy of having no more than two times the average monthly sales in overstock.. Also included is how Melissa and Doug® can switch from a short-term 3D printing manufacturing plan to a more long-term, large-scale injection molding process.

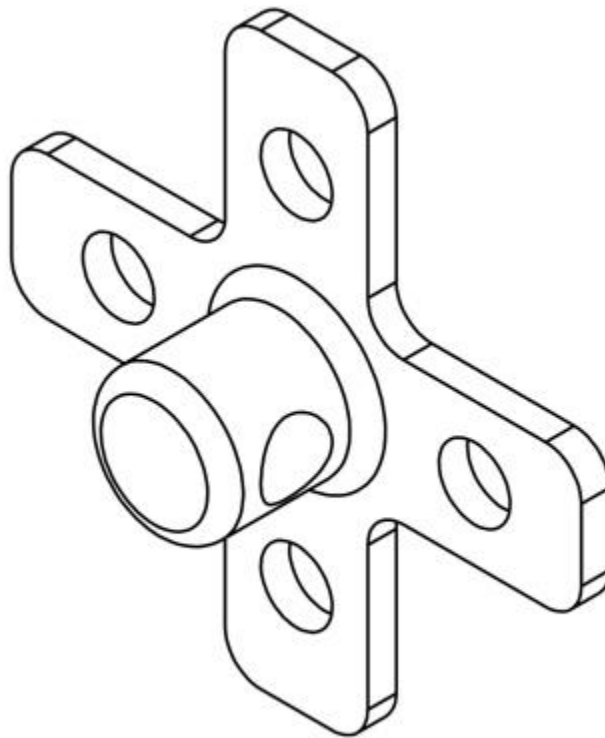


Figure 1: An isometric drawing of the Tower Topper part that must be produced. The piece is 4 in. x 4 in. base

Methods

The manufacturing plan was constructed by using a spreadsheet to keep track of inventory, supply, demand, and profit (Appendix B). Based on the projected sales, 9,000 units needed to be produced from June 2021 to October 2021, which created a constraint of being able to produce 1,800 units per month on average, with 3D printing alone. This is 81 units a day, assuming a 5-day work week, so the 3D printing had to be durable (constraint given by the lead

engineer,) but also fast enough to produce 81 units a day. With 6 Stratasys F170 printers (10x10x10 bed size,) it is possible to print 8 units per bed (Appendix C), and using 3D printing calculator and the .stl file given by the Melissa and Doug® team, it would take 1.49 hours to print one in one bed (Layer Height: 0.28mm, Infill: 20 percent, Print Speed: 30 mm/s, Material: ABS) [1][2]. These inputs were chosen to optimize durability and efficiency. This means that the 3D printing technician, working from 8:00am to 5:00pm, could run a 6-part print on all the printers in 8 hours and 56.4 minutes and then run an 8-part print on all of the printers before he/she leaves. This would result in 84 units per day. The 3D printing cost would need to be covered by Melissa and Doug® in order to afford Injection Molding at the end of Quarter 3 of 2021.

The price of injection molding was calculated using an injection molding calculator, with the inputs hand-calculated based on the given engineering drawing (see Appendix D) [3]. The calculations and input values (with justification) can be seen in (Appendix E.) This resulted in a fixed manufacturing cost of \$12,975 and unit prices of \$1.053, \$1.000, \$0.983, and \$0.974 for 2,500; 5,000; 7,500; and 10,000 units respectively.

Results

The plan is to make just enough parts by 3D-printing in order to afford the creation of the molds at the end of the third fiscal quarter. This is done by printing at full capacity (84 parts a day) every work day in June, July, and August 2021, and for 19 days in September 2021. This gives \$12,996.45 to pay to start the creation of the mold (\$12,975) in the beginning of October 2021. The mold will take one month to make and the first batch of 10,000 parts will take another month to make, and during this time, 3D-printing will return to printing at full capacity every work day. The 3D printing technician must go start 1 print of 8 parts on all of the printers on the first Saturday of October in order to keep up with demand. After October, printing will slow down until February 2022 in order to facilitate the transition between 3D printing and injection molding. The production plan is shown in the timeline below (Figure 2). New injection molds will need to be purchased at the end of May 2022 and the end of October 2022, which is when 40,000 parts have been made from the previous injection molds. The last 10,000 parts will be made from the previous injection mold as the new ones are being constructed. Including the monthly profit margin and the leftover manufacturing money, this manufacturing plan projects to make a profit of \$616,050.60 in the span of the first 18 months, as well as transitions smoothly

from 3D printing into injection molding long term.

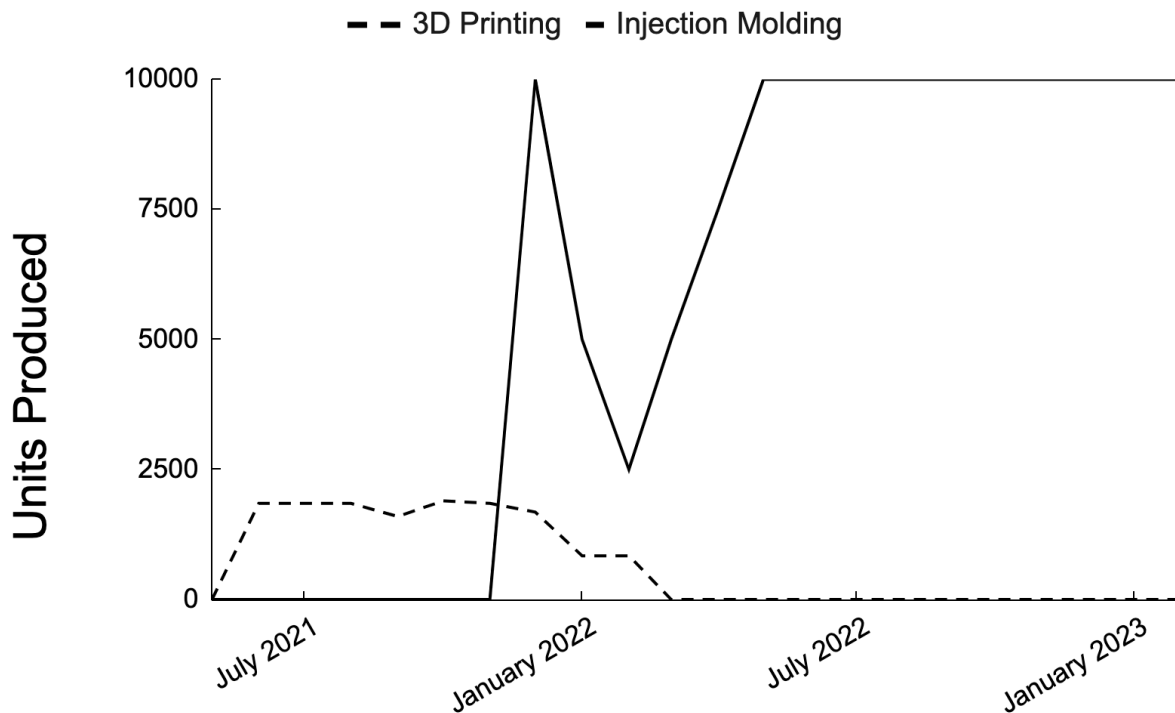


Figure 2: This is a timeline of the production plan for the first 18 months after the Build a Tower product launch. Injection molding will start in the beginning of October 2021, and the first batch will be finished in December. 3D printing will continue until February 2022.

Discussion

The manufacturing plan overall is projected to be successful and was within the constraints given (see Appendix F); however, improvements could be made by increasing the speed at which Melissa and Doug® can 3D print. The time estimate given (1.49 hours per part) is a bit of an overestimate because it does not account for the time saved by putting more than one part on a bed. Increasing the speed of 3D printing could result in more durable parts (Layer height was increased to speed up the process; however, decreasing the layer height would result in a more durable part,) more vacation days for the 3D printing technician, and allow for redundancies in case of print failure. This could be done by increasing the amount of printers or hiring technicians to run the printers 24/7. Money could be saved by shortening the shifts and spreading them out across the day.

References

- [1] “F123 Series: F120, F170, F270, & F370 Professional 3D Printers,” *Stratasys*. [Online]. Available:
<https://www.stratasys.com/3d-printers/f123#:~:text=Stratasys%20F120%20%26%20F170%3A%20254%20x,x%2010%20x%2014%20in.>

- [2] “Price Calculator for 3D Printing,” *3DPartPrice*. [Online]. Available:
<http://3dpartprice.com/>.

- [3] “Cost Estimator,” *Injection Molding Cost Estimator*. [Online]. Available:
<https://www.custompartnet.com/estimate/injection-molding/>.

Appendices

Appendix A: Sales Projections

In order to construct a proper manufacturing plan for the Tower Topper for the first 18 months after the release of the Build a Tower product by Melissa and Doug®, the team provided sales projections based on similar products, shown in Table A.1 below. This was used to project inventory and profit margins for the manufacturing plan.

Table A.1: This shows the projected sales of the Build a Tower for the first 18 months, based on similar products, provided by the Melissa and Doug® team

SALES PROJECTIONS: BUILD A TOWER

Prepared by: Hunter T. Brah, III, Summer Intern in Sales & Marketing

Prepared on: April 1st, 2021

Fiscal Qtr	Year	Month	Units Sold
Q3	2021	Sept	2500
Q4	2021	Oct	3000
Q4	2021	Nov	4500
Q4	2021	Dec	6500
Q1	2022	Jan	3500
Q1	2022	Feb	4100
Q1	2022	Mar	4700
Q2	2022	Apr	5400
Q2	2022	May	6200
Q2	2022	June	7100
Q3	2022	July	8200
Q3	2022	Aug	9400
Q3	2022	Sept	10900
Q4	2022	Oct	12500
Q4	2022	Nov	14400
Q4	2022	Dec	21600
Q1	2023	Jan	6100
Q1	2023	Feb	7000

Appendix B: Inventory Spreadsheet

Table B.1: This is the spreadsheet used to organize the inventory, supply, demand, and profit for the manufacturing plan of the Melissa and Doug® Tower Topper. Total profit was calculated by summing 55% of the gross margin and adding the leftover manufacturing money. White and Gray rows indicate the span of a fiscal quarter. Fixed Manufacturing was paid at the end of the month.

	Projected Sales	Units Produced by 3DP	Units Produced By I.M.	Gross Cost	Units In Stock by the end of the month	Gross Margin	Manufacturing Money	Fixed Manufacturing	Unit Cost For 3DP	Unit Cost for I.M.
May 2021	0	0	0	0	0	0	0	0	4.35	0
Jun 2021	0	1848	0	8038.8	1848	-8038.8	-3617.46	0	4.35	0
July 2021	0	1848	0	8038.8	3696	-8038.8	-7234.92	0	4.35	0
Aug 2021	3500	1848	0	8038.8	2044	26926.2	4881.87	0	4.35	0
September 2021	2500	1596	0	6942.6	1140	18032.4	12996.45	12974	4.35	0
Oct 2021	3000	1896	0	8247.6	36	21722.4	9797.53	0	4.35	0
November 2021	4500	1848	0	8038.8	-2616	36916.2	26409.82	0	4.35	0
Dec 2021	6500	1680	10000	60548	2564	4387	28383.97	0	4.35	5.324
January 2022	3500	840	5000	30404	4904	4561	30436.42	0	4.35	5.35
Feb 2022	4100	840	2500	17161.5	4144	23797.5	41145.295	0	4.35	5.403
March 2022	4700	0	5000	26750	4444	20203	50236.645	0	4.35	5.35
Apr 2022	5400	0	7500	39997.5	6544	13948.5	56513.47	0	4.35	5.333
May 2022	6200	0	10000	53250	10344	8688	60423.07	12974	4.35	5.325
Jun 2022	7100	0	10000	53240	13244	17689	55409.12	0	4.35	5.324
July 2022	8200	0	10000	53240	15044	28678	68314.22	0	4.35	5.324
Aug 2022	9400	0	10000	53240	15644	40666	86613.92	0	4.35	5.324
September 2022	10900	0	10000	53240	14744	55651	111656.87	0	4.35	5.324
Oct 2022	12500	0	10000	53240	12244	71635	143892.62	12974	4.35	5.324
November 2022	14400	0	10000	53240	7844	90616	171695.82	0	4.35	5.324
December 2022	21600	0	10000	53240	-3756	162544	244840.62	0	4.35	5.324
Jan 2023	6100	0	10000	53240	144	7699	248305.17	0	4.35	5.324
February 2023	7000	0	10000	53240	3144	16690	255815.67	0	4.35	5.324

Appendix C: 3D Print Layout

Melissa and Doug® use six Stratasys F170 3D printers to print the Tower Toppers, which can print in a 10x10x10 volume. Figure C.1 shows how to fit 8 tower toppers in a 10x10 bed for printing.

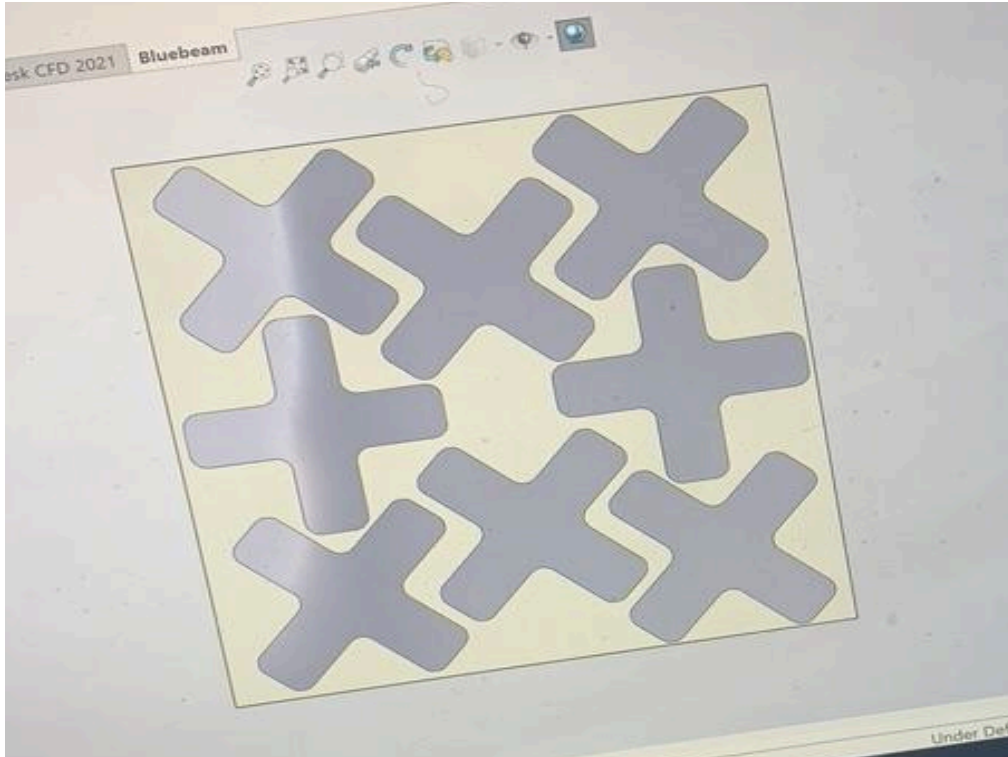


Figure C.1: This shows the layout of the projected base of the Tower Topper such 8 of them can fit on a 10x10 bed

Appendix D: Engineering Drawing

The Melissa and Doug® team provided the engineering drawing of the Tower Topper part of the Build a Tower toy, shown below in Figure D.1. The measurements are used to calculate the area and volume for estimating injection molding costs. The engineering drawing also specifies the required precision.

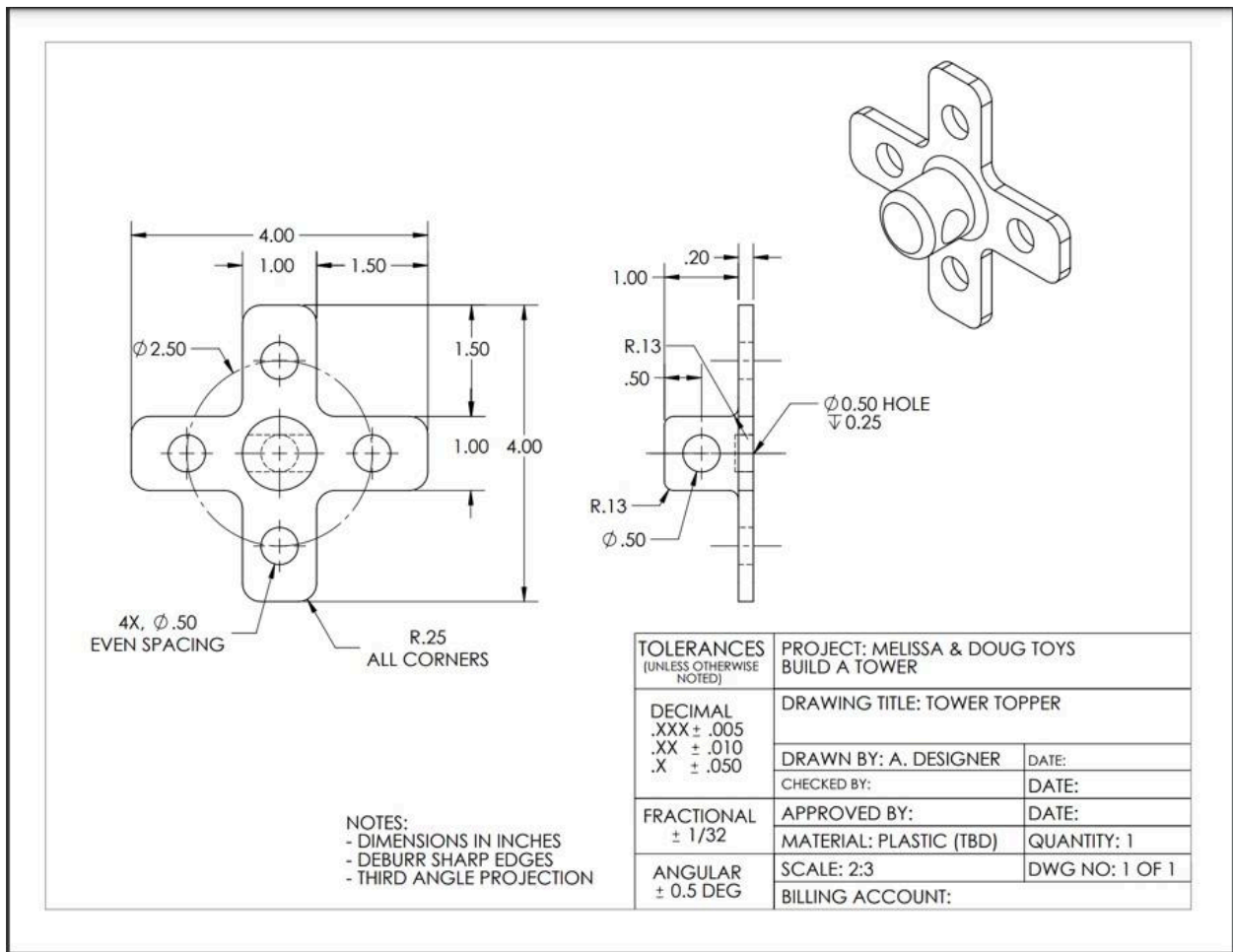


Figure D.1: This is the orthographic and isometric sketch given by the Melissa and Doug® for the Tower Topper part of their Build a Tower product

Appendix E: Inputs and Calculations for Cost Estimators

These are calculations for the input values for the injection molding price calculator for the Melissa and Doug® Tower Topper. Geometry was calculated, shown in (Table E.1), with the input and output values shown in (Figure E.1). Tolerance was chosen to be Moderate Precision based on the tolerances on the given engineering drawing. Smooth was chosen for surface roughness because the company specified that it wanted a matte finish. Very Simple was chosen for Complexity due to the part's small number of features.

Table E.1: This shows the calculations necessary for the inputs for the injection molding price calculator. Measurements were retrieved from the given engineering drawing

Feature	Calculations
Base Area (sq. in.)	$1.5(4) = 7$
Holes (sq. in.)	$4(\pi(0.25)^2) = 0.9817$
Base Volume (cu. in.)	$(7 - 0.9817) * 0.20 = 1.2037$
Cylinder Volume (cu. in.)	$\pi(0.51^2) = 0.7854$
Hole through Cylinder (cu. in)	$\pi(0.25)^2 = 0.1963$
Total Volume (cu. in)	$1.2037 + 0.7854 - 0.1963 = 1.729$

The screenshot displays the 'Injection Molding' software interface, specifically the 'Reports' tab. The 'Part Information' section contains the following inputs and outputs:

- Rapid tooling?**: ☐ Yes ☒ No
- Quantity**: 10000
- Material**: Acrylonitrile Butadiene Styrene (ABS), Molded (with a 'Browse...' button)
- Envelope X-Y-Z (in)**: 4 x 4 x 1.2
- Max. wall thickness (in)**: .25
- Projected area (in²)**: 7 or 43.75 % of envelope
- Projected holes?**: ☒ Yes ☐ No
- Total Area (in²)**: 0.9817 or 6.14 % of envelope
- Volume (in³)**: 1.2037 or 6.27 % of envelope
- Tolerance (in)**: Moderate precision (<= 0.01)
- Surface roughness (µin)**: Smooth (Ra <= 32)
- Complexity**: Very Simple (with a link to 'Show advanced complexity options')

The 'Process Parameters' section is currently empty. The 'Cost' section shows the following breakdown:

- Update Estimate** button
- Material**: \$2,096 (\$0.210 per part)
- Production**: \$7,640 (\$0.764 per part)
- Tooling**: \$12,974 (\$1.297 per part)
- Total**: \$22,710 (\$2.271 per part)

Figure E.1: This shows the input and output values in the injection mold price calculator

Appendix F: Projected Inventory Timeline

In order to make sure that the Tower Topper part did not go into back order for more than a month and did not violate Melissa & Doug's just-in-time manufacturing policy to hold no more than two times the average monthly sales in overstock at any given time, the inventory was kept on a month-by-month basis, shown in Figure F.1 below.

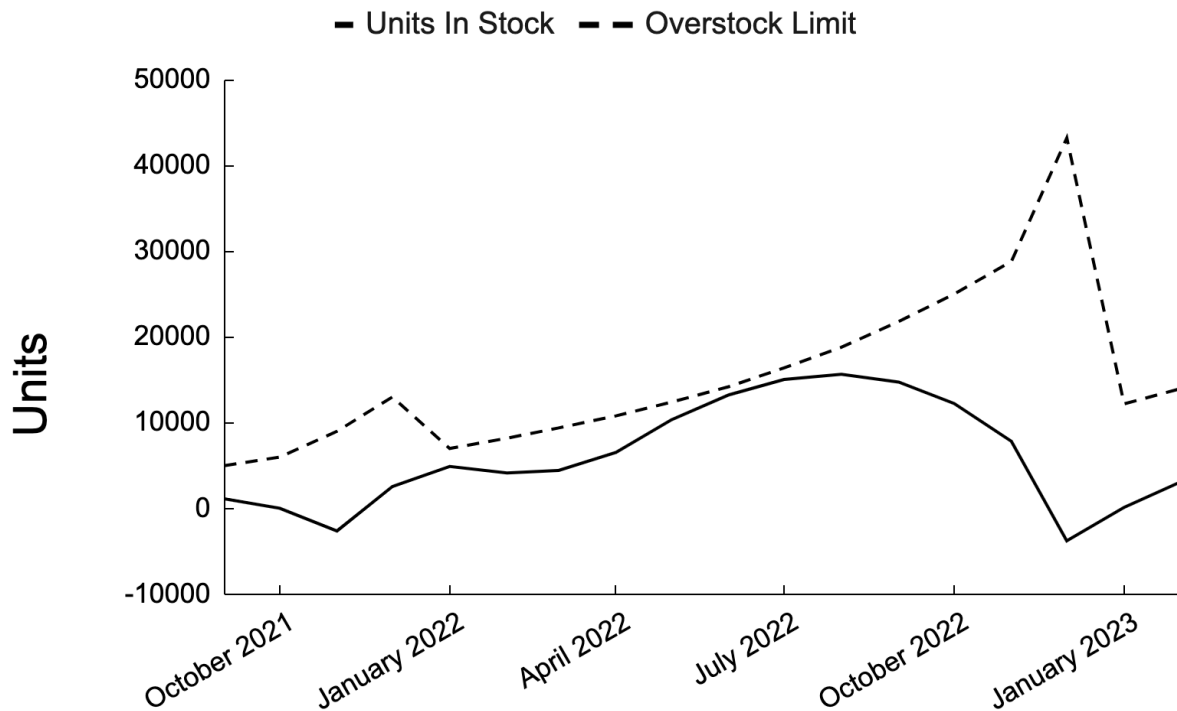


Figure F.1: This is a timeline of the inventory of Tower Toppers, showing that it never goes above the overstock limit and does not stay back ordered for longer than a month.